Ph.D. Student Status Update and Self Evaluation

School of Computing, UGA, Spring 2026

Last Name, First Name: BISHI, Rahman KOREDE

Last four digits of UGA ID: 3422

Semester and Year when you joined the Ph.D. program: Fall, 2022

Date when this form is being submitted: 04/15/2026

Name of the Major Professor: Dr. John A. Miller

Names of the PhD Advisory Committee:

- Committee Member: Dr. John A. Miller (Major Professor)
- Committee Member: Dr. Qianwen Li
- Committee Member: Dr. Ingrid Maria Hybinette
- Committee Member:

MILESTONES

Formation of PhD advisory committee (completion of the G130 Form): Summer, 2025

Core competency: Passed -- July 20, 2025 (Core GPA: 3.566)

Preliminary Doctoral Program of Study Form: Fall, 2023

Paper Submission before Comprehensive Exam: Spring, 2026

- Title: "Beyond Corridor Averages: Lane-Level Validation of Microscopic Freeway Simulation with Data-Driven Arrivals"
- Conference/Journal: Annual Modeling and Simulation Conference (ANNSIM 2026)

Final Program of Study Form (G138): NM (Spring, 2026)

Comprehensive Exam:

- Written Exam: In progress -- Spring, 2026
- Oral Exam: NM (Summer, 2026)

Application for Admission to Candidacy: NM (Summer, 2026)

Paper Accepted Before Prospectus: Spring, 2026

- Title: "Beyond Corridor Averages: Lane-Level Validation of Microscopic Freeway Simulation with Data-Driven Arrivals"
- Conference/Journal: Annual Modeling and Simulation Conference (ANNSIM 2026) -- Accepted

Prospectus: NM (Fall, 2026)

Application for Graduation Form: NM (Spring, 2028)

Defense: NM (Spring, 2028)

RESEARCH PROGRESS

Research Progress Summary

My dissertation focuses on microscopic traffic simulation using the ScalaTion 2.0 framework. Over the past year I built a full simulation engine for the US-101 freeway corridor in the San Francisco Bay Area, validated against lane-level sensor data from Caltrans PeMS. I implemented three car-following models (IDM, Gipps, Krauss), a MOBIL lane-change model, on-ramp merge behavior, and a multi-lane doubly-linked-list vehicle tracking structure supporting variable lane counts. I ran calibration experiments on the GACRC Sapelo2 cluster using SPSA and Nelder-Mead optimizers. My first paper was accepted at ANNSIM 2026, showing that arrival-process choice has far greater impact on simulation accuracy than numerical integrator choice -- the shifted Erlang-2 distribution reduced flow prediction error by roughly 28% compared to Poisson. I am currently preparing for my written candidacy examination.

List of Publications

1. Bishi, K.R., Bowman, J., Miller, J.A. (2026). "Beyond Corridor Averages: Lane-Level Validation of Microscopic Freeway Simulation with Data-Driven Arrivals." Annual Modeling and Simulation Conference (ANNSIM 2026). [Accepted]

COURSE WORK PROGRESS

Number of credit hours completed by end of current semester:

Number of credit hours completed by end of current semester that count towards program of study:

Coursework completed or to be completed by end of this semester:

No. of hours completed from preliminary focus by end of this semester:

No. of hours completed from the primary focus area by the end of this semester:

No. of hours of 8000-level CSCI coursework:

No. of hours of graduate-level CSCI coursework:

No. of hours of doctoral minor coursework (if doing a minor): None

No. of hours of CSCI 9300 Doctoral Dissertation:

ACTIVITIES DURING CURRENT YEAR (Summer 2025, Fall 2025 and Spring 2026)

Research Activities During this Academic Year Including Publications

- Designed and built a microscopic traffic simulation of the US-101 freeway corridor (Willow Road to Marsh Road, ~2 miles) within the ScalaTion 2.0 discrete-event simulation framework
- Implemented three car-following dynamics -- Intelligent Driver Model (IDM), Gipps safe-distance model, and Krauss stochastic model -- with configurable numerical integrators (Euler through Dormand-Prince RK45)
- Built lane-level validation infrastructure comparing simulated per-lane flow and speed against PeMS sensor data at five detector stations
- Developed a shifted Erlang-2 arrival process that enforces realistic minimum headways, reducing flow prediction error by ~28% versus standard Poisson arrivals
- Implemented MOBIL lane-change model (Treiber & Kesting, 2007) within the simulation engine, including safety criterion, incentive criterion, and anti-oscillation cooldown
- Built variable-lane doubly-linked-list architecture supporting segments with different lane counts (e.g., 4-lane mainline with 2-lane on-ramps)
- Ran calibration experiments on the GACRC Sapelo2 HPC cluster using SPSA optimizer with SLURM array jobs
- Built a second simulation model for the I-210 / SR-134 corridor (Eaton Fire evacuation study, dual-corridor with on-ramps and off-ramps)
- Developed a rendering engine using Java2D: road polygons, lane markings, rotated velocity-colored vehicles, OpenStreetMap background map overlay with auto-download, HUD overlay
- Paper accepted at ANNSIM 2026: "Beyond Corridor Averages: Lane-Level Validation of Microscopic Freeway Simulation with Data-Driven Arrivals"

Course-related Activities During this Academic Year

(Fill in your Summer 2025, Fall 2025, Spring 2026 courses here -- example format:)

- CSCI 9000 -- Summer 2025
- CSCI 9000 -- Fall 2025
- CSCI 9000 -- Spring 2026

Degree-related Milestones Achieved During This Academic Year

- Core Competency certified -- July 20, 2025 (Core GPA 3.566, approved by full committee)
- PhD Advisory Committee formed -- Summer, 2025 (Dr. Miller, Dr. Li, Dr. Hybinette)
- First paper submitted and accepted -- ANNSIM 2026
- Written candidacy examination -- in progress (Spring, 2026)

NOTABLE ACHIEVEMENTS/RECOGNITIONS YOU MAY HAVE HAD DURING THIS YEAR

None

DEVIATIONS FROM PLAN THAT YOU PROVIDED LAST YEAR

Last year I planned to submit a paper on "Dual Training of Foundation Models Using Simulation and Digital Twins" to IEEE BigData. My research direction shifted after closer work with Dr. Miller toward microscopic traffic simulation and empirical validation against PeMS sensor data. The paper I ended up writing and getting accepted at ANNSIM 2026 addresses lane-level validation of freeway simulation with data-driven arrivals -- a more focused and empirically grounded topic than the original plan. The core competency form, which was completed but not yet submitted last year, was formally signed and certified in July 2025. The committee was also formed this year, which was on track with my estimated timeline from last year (Fall 2024 estimate, completed Summer 2025).

PLANS FOR THE NEXT YEAR

Research-related Plans

I plan to complete my written and oral candidacy examinations by Summer 2026. My next research target is a paper on wildfire evacuation resilience, using the I-210 corridor simulation I have already built to evaluate contraflow strategies during the 2025 Palisades Fire. I will calibrate the model against PeMS fire-day sensor data and run counterfactual scenarios comparing baseline, partial contraflow, and full contraflow configurations. I also plan to continue improving the simulation engine -- gap acceptance at ramp merges, density-based lane assignment, and further calibration work on Sapelo2. I expect to submit one more paper by end of 2026.

Course Work-Related Plans

I have completed most of my required coursework. Unless directed otherwise by my major professor, I do not plan on taking additional courses beyond research hours (CSCI 9000/9300).

Degree-related Milestones

- Complete written candidacy exam -- Spring 2026
- Complete oral candidacy exam -- Summer 2026
- Admission to candidacy -- Summer 2026
- Submit second paper -- Fall 2026
- Prospectus -- Fall 2026 or Spring 2027

FUNDING HISTORY

Semester and Year	Funding Type	Unit Providing Funding	If funded as RA, faculty name(s)
Fall 2022	TA	SoC	
Spring 2023	TA	SoC	
Summer 2023	TA	SoC	
Summer 2023	RA	EETI	Dr. Nathaniel Hunsu

Fall 2023	TA	SoC	
Spring 2024	TA	SoC	
Summer 2024	(fill in)	(fill in)	
Fall 2024	(fill in)	(fill in)	
Spring 2025	(fill in)	(fill in)	
Summer 2025	(fill in)	(fill in)	
Fall 2025	(fill in)	(fill in)	
Spring 2026	(fill in)	(fill in)	

SIGNATURES:

Signature of Student: Date:

Signature of Major Professor: Date: